Anuj Das

AI and ML Engineer

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AI and ML engineering student (graduating 2027) building computer vision, sensor-fusion and edge systems that work outside the lab: industrial IoT software at Larsen & Toubro, an offline rescue drone, and full-stack AI products.

EXPERIENCE

Software Engineering Intern · Larsen & Toubro (L&T), Chennai — Jun 2025 – Jul 2025
Python · Scikit-learn · React.js · IoT

- Built a Scikit-learn pipeline that classifies and predicts sensor anomalies in industrial chillers, cutting fault detection time by ~40%, and an ML-driven alert system that cut failure response from hours to under 5 minutes.
- Engineered a real-time IoT ingestion layer for 10+ live devices, feeding the operations team's daily React.js dashboard.
- Developed a digital twin of the chiller network: a React.js plant map showing 20+ valves in real time, with virtual valve control and temperature, pressure and flow data at under 2-second latency, reducing operator intervention by ~60%.
- Shipped production-ready software in a 45-day internship with mechanical and electrical engineers across 3 teams.

PROJECTS

SkyNetics Rescue Drone · Team of 4, software — 2026
Python · YOLOv8 · ONNX Runtime · OpenCV · Raspberry Pi 4 · Sensor fusion · GitHub

- Built the on-device software for a fully offline drone that locates people buried by avalanches and landslides by fusing thermal imaging, mmWave radar and YOLOv8 vision on a Raspberry Pi 4.
- Designed a three-thread capture, inference and display pipeline that hands off only the latest frame, keeping the pilot's live overlay at ~30 fps even when inference stalls.
- Implemented adaptive thermal thresholding, confidence-weighted sensor fusion and Kalman + SORT multi-object tracking.

Shigotai · AI career platform for jobs in Japan — 2026
Next.js · TypeScript · FastAPI · SQLAlchemy · LLM · Live demo · GitHub

- Built a full-stack platform that matches candidates to live openings at Japanese companies, ingesting postings from the Greenhouse, Lever, Ashby and SmartRecruiters APIs.
- Designed a three-stage matching engine (rule-based eligibility, explainable semantic matching, LLM explanations only for shortlisted jobs) and a resume parser that verifies every extracted field against the source text.

SmartInvoice AI · Document intelligence system — Jan 2026 – Feb 2026
Python · OCR · Machine Learning · NLP

- Built an OCR pipeline that extracts 15+ structured fields from unstructured PDF invoices with 89% field-level accuracy, reducing processing time by 70% versus manual entry.
- Added explainable per-field confidence scores so auditors review only low-confidence extractions.

EdgeGrid · Firmware management for edge devices — 2026
React · TypeScript · Django REST · PostgreSQL · Redis · Celery · Docker · GitHub

- Built a control plane for heterogeneous edge devices with per-device-type firmware compatibility, verified installs, progress reporting and automatic rollback on failure.
- Live device health over Django Channels WebSockets, background jobs on Celery and Redis, packaged with Docker.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
ML and Computer Vision: Scikit-learn, TensorFlow, YOLOv8, ONNX Runtime, OpenCV, Pandas, NumPy, OCR
Web and Backend: React, Next.js, Node.js, FastAPI, Django REST Framework, HTML5, CSS3, Tailwind CSS
Data and DevOps: PostgreSQL, MySQL, Redis, Celery, Docker, Git
Edge and IoT: Raspberry Pi, sensor fusion, thermal imaging, mmWave radar, real-time telemetry
Spoken languages: English, Hindi, Japanese (JLPT N3)

EDUCATION

B.E. Computer Science Engineering (AI and ML) · Easwari Engineering College, Chennai — CGPA 8.63 · 2023 – 2027

Class I to XII · Kendriya Vidyalaya No. 1, Mangalore — 2010 – 2023

CERTIFICATIONS

Oracle Certified Professional: Java SE 11 Developer · Oracle, 2026
IoT and Privacy in Social Media · NPTEL, 2025
JLPT N3 · Japanese-Language Proficiency Test
NAT-TEST 3Q · Japanese language proficiency